

Practical File
On
DATABASE MANAGEMENT SYSTEMS

Submitted in partial fulfilment of the requirements for
the award of the degree

Bachelor of Technology
in
Department of Computer Science and Engineering



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Ex – 1

Employee Directory

1. Aim

To design a relational schema for an employee directory with departments and employees, enforce primary-key and foreign-key constraints, create a reporting view, and verify the design with sample queries in SQLite.

2. Theory

2.1 Tables and keys

A departments table stores each organisational unit. An employees table stores people; department_id is a foreign key to departments, and manager_id is a self-referencing foreign key for the reporting hierarchy.

2.2 Views

A view (employee_directory) joins employees to their department and manager names so reporting queries do not repeat join logic.

3. Schema

Table	Key columns	Role
departments	department_id PK	Organisational units
employees	employee_id PK; department_id FK	Staff records
employees	manager_id FK → employees	Reporting chain
employee_directory	VIEW	Flattened directory for queries

4. Procedure

1. Create departments and employees with constraints and indexes.
2. Create the employee_directory view.
3. Insert six departments and twenty-six employees with a three-tier org chart.
4. Run aggregate queries on the view (counts by department).

5. Source Code

File: 19-08-2026/schema.sql

```
-- =====
-- Employee Directory – schema
-- SQLite dialect. Runs as-is in the in-browser SQL Compiler
-- (index.html), and in the sqlite3 CLI / DB Browser for SQLite.
-- =====

PRAGMA foreign_keys = ON;

DROP VIEW IF EXISTS employee_directory;
DROP TABLE IF EXISTS employees;
DROP TABLE IF EXISTS departments;

-- -----
-- departments
-- -----
CREATE TABLE departments (
    department_id INTEGER PRIMARY KEY AUTOINCREMENT,
    name          TEXT NOT NULL UNIQUE,
```

```

        description    TEXT,
        color_hex      TEXT NOT NULL DEFAULT '#1CB0F6'
    );

-- -----
-- employees
-- -----
CREATE TABLE employees (
    employee_id        INTEGER PRIMARY KEY AUTOINCREMENT,
    first_name         TEXT NOT NULL,
    last_name          TEXT NOT NULL,
    email              TEXT NOT NULL UNIQUE,
    job_title           TEXT NOT NULL,
    department_id      INTEGER NOT NULL REFERENCES departments (department_id) ON DELETE RESTRICT,
    manager_id         INTEGER REFERENCES employees (employee_id) ON DELETE SET NULL,
    salary             NUMERIC(10,2) NOT NULL CHECK (salary > 0),
    hire_date           TEXT NOT NULL,
    employment_status  TEXT NOT NULL DEFAULT 'active'
                    CHECK (employment_status IN ('active', 'remote', 'on_leave')),
    avatar_seed        TEXT NOT NULL
);

CREATE INDEX idx_employees_department ON employees (department_id);
CREATE INDEX idx_employees_manager   ON employees (manager_id);

-- -----
-- employee_directory – flattened view joining department + manager.
-- Convenient for reporting queries and reused by the frontend.
-- -----
CREATE VIEW employee_directory AS
SELECT
    e.employee_id,
    e.first_name || ' ' || e.last_name AS full_name,
    e.email,
    e.job_title,
    d.name                               AS department_name,
    d.color_hex                         AS department_color,
    m.first_name || ' ' || m.last_name AS manager_name,
    e.salary,
    e.hire_date,
    e.employment_status,
    e.avatar_seed
FROM employees e
JOIN departments d ON d.department_id = e.department_id
LEFT JOIN employees m ON m.employee_id = e.manager_id;

```

6. Output

```

SQL> SELECT COUNT(*) AS dept_count FROM departments;
dept_count
6

```

```

SQL> SELECT COUNT(*) AS emp_count FROM employees;
emp_count
26

```

```

SQL> SELECT department_name, COUNT(*) AS headcount
FROM employee_directory
GROUP BY department_name
ORDER BY headcount DESC;
department_name headcount
Engineering      8
Design           5
Marketing        4
Sales            4
People Ops       3
Finance          2

```

```

SQL> SELECT full_name, department_name, manager_name
FROM employee_directory
LIMIT 5;
full_name      department_name manager_name
Sam Okafor     Engineering    NULL
Priya Nair     Engineering    Sam Okafor
Elena Petrova  Design        Sam Okafor
Grace Adeyemi  Marketing     Sam Okafor
Isabella Rossi Sales          Sam Okafor

```

7. Results

Six departments and twenty-six employees load successfully. Foreign keys prevent orphan department_id values. The view returns each employee with department and manager names for browsing and SQL practice.

8. Conclusion

Normalised tables plus a view provide a maintainable employee directory. Primary and foreign keys preserve referential integrity; the view simplifies read-only reporting queries used in the web compiler.

Ex – 2

CREATE & ALTER TABLE

1. Aim

To create an employees table with attributes empID, empName, department and salary, insert sample records, and then rename the column department to faculty using ALTER TABLE.

2. Theory

2.1 CREATE TABLE

A relational table is defined by a name, a list of attributes, and constraints that protect the data. PRIMARY KEY makes empID unique and not null. NOT NULL rejects missing names and faculty values. CHECK (salary > 0) rejects non-positive salaries. NUMERIC(10, 2) stores pay as a decimal with two fractional digits.

2.2 ALTER TABLE RENAME COLUMN

In SQLite and MySQL 8+, ALTER TABLE ... RENAME COLUMN changes a column's name without rewriting the stored values. After RENAME COLUMN department TO faculty, every row still holds the same school name; only the heading of that column becomes faculty. This is a schema change, not an UPDATE. Microsoft SQL Server does not accept that syntax; the equivalent is EXEC sp_rename 'employees.department', 'faculty', 'COLUMN' (see employees.sqlserver.sql).

3. Schema

The table is created with four attributes. After the rename, the third attribute is faculty instead of department.

Attribute	Type	Constraint	Role
empID	INTEGER	PRIMARY KEY	Employee identifier
empName	TEXT	NOT NULL	Employee name
department → faculty	TEXT	NOT NULL	School / faculty name
salary	NUMERIC(10,2)	NOT NULL, CHECK > 0	Monthly salary

Sample faculty values include **School of Engineering and Technology** (the faculty named on this report's cover). The cover also lists **Department of CSE** as the student's home department; that heading is letterhead, not a fifth column in the table.

4. Procedure

1. Drop employees if it already exists, so the script can be re-run.
2. Create the table with empID, empName, department and salary.
3. Insert six sample employees.
4. Display all rows and the column list with PRAGMA table_info.
5. Rename department to faculty.
6. Display all rows and the column list again to confirm that only the name changed.

5. Source Code

```
-- =====
-- DBMS Lab · 26-08-2026
-- Employees table: CREATE, seed, then RENAME department → faculty
```

```
-- SQLite (sql.js compiler / DB Browser / sqlite3)
-- For the class SQL Server, run employees.sqlserver.sql instead.
-- =====

DROP TABLE IF EXISTS employees;

-- -----
-- 1. Create the employees table
-- -----
CREATE TABLE employees (
    empID      INTEGER PRIMARY KEY,
    empName    TEXT NOT NULL,
    department TEXT NOT NULL,
    salary     NUMERIC(10, 2) NOT NULL CHECK (salary > 0)
);

-- -----
-- 2. Insert sample rows (department holds the faculty/school name)
-- -----
INSERT INTO employees (empID, empName, department, salary) VALUES
    (1, 'Ananya Sharma', 'School of Engineering and Technology', 92000.00),
    (2, 'Rohan Mehta', 'School of Management', 78000.00),
    (3, 'Priya Nair', 'School of Law', 85000.00),
    (4, 'Vikram Singh', 'School of Agricultural Sciences', 76000.00),
    (5, 'Fatima Khan', 'School of Engineering and Technology', 88000.00),
    (6, 'Arjun Patel', 'School of Medical Sciences', 95000.00);

-- -----
-- 3. Verify the table BEFORE the rename
-- -----
SELECT * FROM employees;
PRAGMA table_info(employees);

-- -----
-- 4. Rename the column department → faculty
-- Row values are unchanged; only the column name changes.
-- -----
ALTER TABLE employees RENAME COLUMN department TO faculty;

-- -----
-- 5. Verify the table AFTER the rename
-- -----
SELECT * FROM employees;
PRAGMA table_info(employees);
```

6. Output

The script was executed in SQLite. PRAGMA table_info columns are cid (column index), name, type, notnull, dflt_value and pk.

```
SQL> DROP TABLE IF EXISTS employees;
-- OK

SQL> CREATE TABLE employees (
    empID      INTEGER PRIMARY KEY,
    empName    TEXT NOT NULL,
    department TEXT NOT NULL,
    salary     NUMERIC(10, 2) NOT NULL CHECK (salary > 0)
);
-- OK

SQL> INSERT INTO employees (empID, empName, department, salary) VALUES
    (1, 'Ananya Sharma', 'School of Engineering and Technology', 92000.00),
    (2, 'Rohan Mehta', 'School of Management', 78000.00),
    (3, 'Priya Nair', 'School of Law', 85000.00),
    (4, 'Vikram Singh', 'School of Agricultural Sciences', 76000.00),
    (5, 'Fatima Khan', 'School of Engineering and Technology', 88000.00),
    (6, 'Arjun Patel', 'School of Medical Sciences', 95000.00);
-- 6 row(s) inserted

SQL> SELECT * FROM employees;
empID empName      department                                salary
-----
1     Ananya Sharma School of Engineering and Technology      92000
2     Rohan Mehta   School of Management                     78000
3     Priya Nair    School of Law                           85000
4     Vikram Singh  School of Agricultural Sciences           76000
5     Fatima Khan   School of Engineering and Technology      88000
6     Arjun Patel   School of Medical Sciences                95000

SQL> PRAGMA table_info(employees);
cid  name      type      notnull  dflt_value  pk
----
0    empID     INTEGER   0        0            1
```

```

1  empName  TEXT          1          0
2  department TEXT        1          0
3  salary   NUMERIC(10, 2) 1          0

SQL> ALTER TABLE employees RENAME COLUMN department TO faculty;
-- OK

SQL> SELECT * FROM employees;
empID  empName      faculty                                     salary
-----
1      Ananya Sharma School of Engineering and Technology      92000
2      Rohan Mehta   School of Management                     78000
3      Priya Nair    School of Law                           85000
4      Vikram Singh  School of Agricultural Sciences          76000
5      Fatima Khan   School of Engineering and Technology     88000
6      Arjun Patel   School of Medical Sciences               95000

SQL> PRAGMA table_info(employees);
cid  name      type      notnull  dflt_value  pk
-----
0    empID     INTEGER   0        1           1
1    empName   TEXT      1        0           0
2    faculty   TEXT      1        0           0
3    salary    NUMERIC(10, 2) 1        0           0

```

7. Results

Stage	Third column	Row count	Values changed?
Before ALTER	department	6	No — original schema
After ALTER	faculty	6	No — names only

All six employees remain. School names such as School of Engineering and Technology are unchanged. The schema listing is the proof: `cid = 2` is named `department` before the statement and `faculty` afterwards.

8. Conclusion

An employees relation can be created with a primary key, not-null attributes and a salary check, then adjusted with `ALTER TABLE ... RENAME COLUMN`. Renaming `department` to `faculty` updates the schema only. The stored values do not change, which is the correct way to relabel an attribute without rewriting the table.

Ex – 3

SQL Joins

1. Aim

To combine rows from related tables using `CROSS JOIN`, `NATURAL JOIN`, `INNER JOIN`, `LEFT OUTER JOIN`, `RIGHT OUTER JOIN`, and `SELF JOIN`; to compare how each join type treats unmatched rows; and to understand when to use each one.

2. Theory

2.1 CROSS JOIN

A `CROSS JOIN` returns the Cartesian product of two tables: every row from the first table is paired with every row from the second. There is no join condition. With 6 departments and 7 employees the result has $6 \times 7 = 42$ **rows**. Cross joins are rare in everyday queries but useful for generating combinations (e.g. all size–colour pairs) or as a building block for other joins.

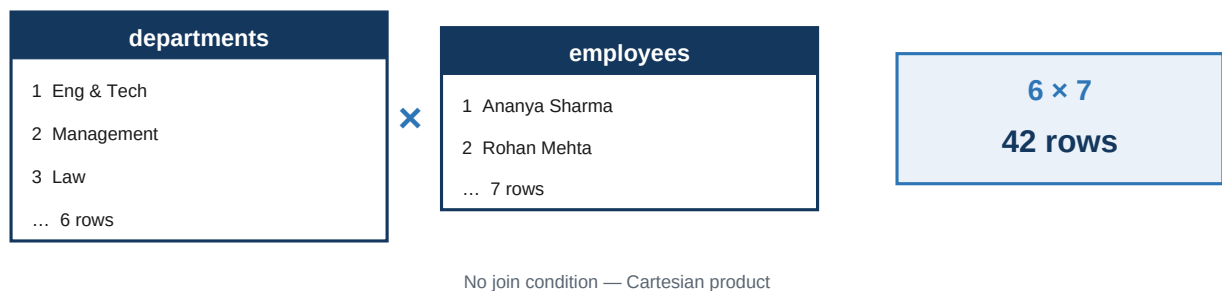


Figure 1 — `CROSS JOIN` pairs every department with every employee (42 combinations).

2.2 NATURAL JOIN

A `NATURAL JOIN` automatically matches rows on every column that appears in both tables with the same name. Here, both tables share `deptID`, so the condition is implicit. Only rows with a matching `deptID` in both tables are returned — the same six rows as `INNER JOIN` when `deptID` is the sole common column. (SQL Server does not support `NATURAL JOIN`; use an explicit `INNER JOIN` instead.)

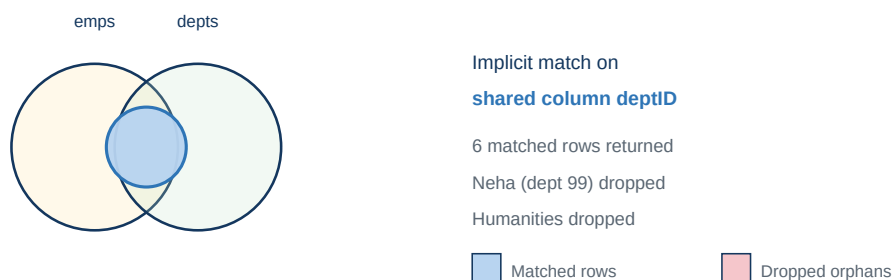


Figure 2 — `NATURAL JOIN` keeps only rows where `deptID` matches in both tables.

2.3 INNER JOIN

An **INNER JOIN** returns only rows where the explicit join condition is satisfied. Unmatched rows from either table are discarded. It is the most common join: use it when you only want records that have a partner on both sides.

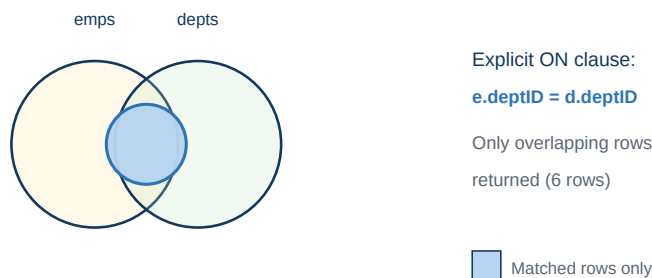


Figure 3 — INNER JOIN returns only the overlapping region (6 matched rows).

2.4 LEFT OUTER JOIN

A **LEFT OUTER JOIN** keeps every row from the left table. If no match exists in the right table, the right-hand columns are filled with **NULL**. Employee Neha Gupta (deptID 99) has no matching department and therefore appears with a null department name.

employees (left table)	LEFT JOIN result
Ananya dept=1	Ananya Eng & Tech
Rohan dept=2	Rohan Management
Priya dept=3	Priya Law
Neha dept=99 ← no match	Neha NULL

All left rows kept — unmatched right columns print as NULL

Figure 4 — LEFT OUTER JOIN preserves Neha (deptID 99); deptName is NULL.

2.5 RIGHT OUTER JOIN

A **RIGHT OUTER JOIN** keeps every row from the right table. Unmatched left-hand columns become **NULL**. School of Humanities (deptID 6) has no employees and therefore appears with null employee fields. A **RIGHT JOIN** can always be rewritten as a **LEFT JOIN** by swapping the table order.

departments (right table)	RIGHT JOIN result
1 Eng & Tech	Ananya Eng & Tech
2 Management	Rohan Management
3 Law	Priya Law
6 Humanities ← no employees	NULL Humanities

All right rows kept — unmatched left columns print as NULL

Figure 5 — RIGHT OUTER JOIN preserves Humanities (deptID 6); employee columns are NULL.

2.6 SELF JOIN

A SELF JOIN joins a table to itself using two different aliases. It is not a separate SQL keyword — it is an INNER or LEFT JOIN where both sides are the same table. Here, employees e is joined to employees m on e.managerID = m.empID to list each employee alongside their manager's name.

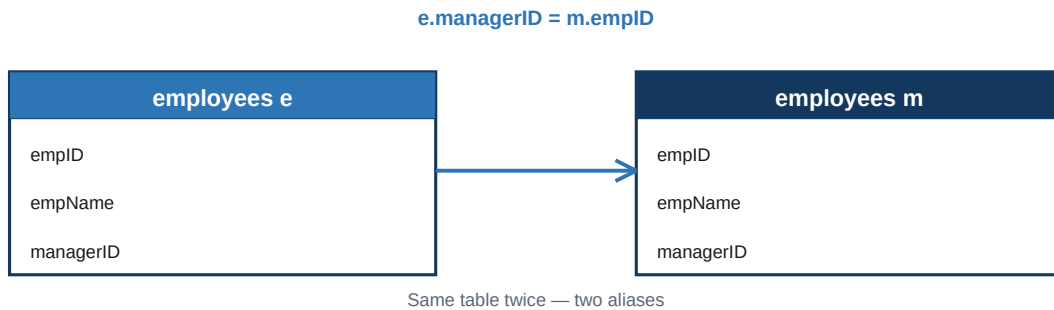


Figure 6 — SELF JOIN: the same table appears twice with different aliases.

2.7 Differences between all join types

Join type	Condition	Unmatched rows	Typical use
CROSS JOIN	None (Cartesian product)	N/A — all combinations	Generate all pairs
NATURAL JOIN	Implicit (shared names)	Dropped from both sides	Quick join on a shared key
INNER JOIN	Explicit ON clause	Dropped from both sides	Only matching records
LEFT OUTER JOIN	Explicit ON clause	Left kept; right NULL	Keep all from the left table
RIGHT OUTER JOIN	Explicit ON clause	Right kept; left NULL	Keep all from the right table
SELF JOIN	Explicit ON (same table)	Depends on INNER / LEFT	Hierarchies, comparisons

Key distinctions: CROSS JOIN multiplies rows with no filter. INNER and NATURAL return only matches. LEFT and RIGHT OUTER preserve one side's orphans. SELF JOIN is a technique (same table, two aliases), not a separate result category — it uses inner or outer join semantics on a single table.

3. Schema

Two tables are linked by deptID. employees also references itself through managerID for the self join (see Figure 6).

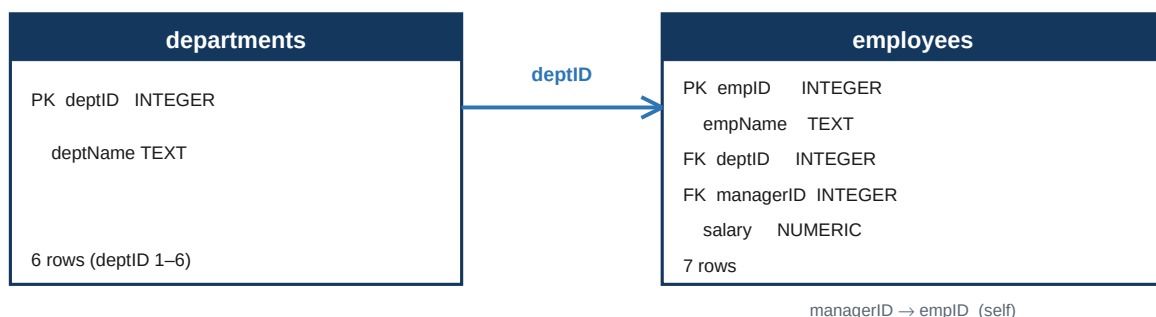


Figure 7 — Schema: employees.deptID references departments.deptID; managerID is a self-reference.

Table	Attribute	Type	Role
departments	deptID	INTEGER / INT	Primary key
departments	deptName	TEXT / VARCHAR	School name
employees	empID	INTEGER / INT	Primary key
employees	empName	TEXT / VARCHAR	Employee name
employees	deptID	INTEGER / INT	FK → departments
employees	managerID	INTEGER / INT	FK → employees (self)
employees	salary	NUMERIC(10,2)	Monthly salary

4. Procedure

1. Create departments and employees (with managerID for self join).
2. Insert six departments and seven employees, including deliberate mismatches for outer joins.
3. Display both base tables.
4. Run `CROSS JOIN` — expect 42 rows (count shown; sample limited to 8).
5. Run `NATURAL JOIN` and `INNER JOIN` — expect six matched rows.
6. Run `LEFT OUTER JOIN` — expect seven rows (one employee without a department).
7. Run `RIGHT OUTER JOIN` — expect seven rows (one department without employees).
8. Run `SELF JOIN` — expect seven rows listing each employee and their manager.

5. Source Code

SQLite script used in the lab compiler (02-09-2026/joins.sql). For the class SQL Server, run joins.sqlserver.sql.

```
-- =====
-- DBMS Lab · 02-09-2026
-- SQL Joins: CROSS, NATURAL, INNER, OUTER (LEFT/RIGHT), SELF
-- SQLite (sql.js compiler / DB Browser / sqlite3)
-- For the class SQL Server, run joins.sqlserver.sql instead.
-- =====

DROP TABLE IF EXISTS employees;
DROP TABLE IF EXISTS departments;

-----
-- 1. Create related tables
--   · deptID links employees → departments
--   · managerID links employees → employees (for SELF JOIN)
-----
CREATE TABLE departments (
    deptID    INTEGER PRIMARY KEY,
    deptName  TEXT NOT NULL
);

CREATE TABLE employees (
    empID     INTEGER PRIMARY KEY,
    empName   TEXT NOT NULL,
    deptID    INTEGER NOT NULL,
    managerID INTEGER,
    salary    NUMERIC(10, 2) NOT NULL CHECK (salary > 0)
);

-----
-- 2. Insert sample rows
--   · deptID 6 (Humanities) has no employees → RIGHT OUTER JOIN
--   · empID 7 points to deptID 99 (missing) → LEFT OUTER JOIN
--   · managerID links form a simple reporting chain → SELF JOIN
-----
INSERT INTO departments (deptID, deptName) VALUES
    (1, 'School of Engineering and Technology'),
    (2, 'School of Management'),
    (3, 'School of Law'),
    (4, 'School of Agricultural Sciences'),
    (5, 'School of Medical Sciences'),
    (6, 'School of Humanities');

INSERT INTO employees (empID, empName, deptID, managerID, salary) VALUES
    (1, 'Ananya Sharma', 1, NULL, 92000.00),
    (2, 'Rohan Mehta', 2, 1, 78000.00),
    (3, 'Priya Nair', 3, 1, 85000.00),
    (4, 'Vikram Singh', 4, 2, 76000.00),
    (5, 'Fatima Khan', 1, 1, 88000.00),
    (6, 'Arjun Patel', 5, 3, 95000.00),
    (7, 'Neha Gupta', 99, 2, 54000.00);

-----
-- 3. Base tables (reference)
-----
-- List all departments
SELECT * FROM departments;
-- List all employees
SELECT * FROM employees;

-----
-- 4. CROSS JOIN
--   Cartesian product: every row of table A paired with every
--   row of table B. No join condition. Here: 6 × 7 = 42 rows.
-----
-- Count cross join rows
SELECT COUNT(*) AS cross_join_row_count
FROM departments
CROSS JOIN employees;

-- Cross join sample pairs
SELECT d.deptName, e.empName
FROM departments AS d
CROSS JOIN employees AS e
ORDER BY d.deptID, e.empID
LIMIT 8;

-----
-- 5. NATURAL JOIN
--   Matches on every column with the same name (here: deptID).
--   Only rows with a matching deptID in BOTH tables appear.
-----
```

```

-- Natural join on deptID
SELECT empID, empName, deptName, salary
FROM employees
NATURAL JOIN departments;

-----
-- 6. INNER JOIN
--   Explicit join condition; same result as NATURAL JOIN here
--   because deptID is the only common column.
-----
-- Inner join employees and departments
SELECT e.empID, e.empName, d.deptName, e.salary
FROM employees AS e
INNER JOIN departments AS d ON e.deptID = d.deptID;

-----
-- 7a. LEFT OUTER JOIN
--   All rows from the LEFT table (employees); unmatched dept
--   columns are NULL (empID 7 → deptID 99 has no department).
-----
-- Left join – keep all employees
SELECT e.empID, e.empName, e.deptID, d.deptName, e.salary
FROM employees AS e
LEFT OUTER JOIN departments AS d ON e.deptID = d.deptID;

-----
-- 7b. RIGHT OUTER JOIN
--   All rows from the RIGHT table (departments); unmatched
--   employee columns are NULL (deptID 6 has no employees).
--   Requires SQLite 3.39+; see joins.sqlserver.sql for T-SQL.
-----
-- Right join – keep all departments
SELECT e.empID, e.empName, d.deptID, d.deptName, e.salary
FROM employees AS e
RIGHT OUTER JOIN departments AS d ON e.deptID = d.deptID;

-----
-- 8. SELF JOIN
--   A table joined to itself using different aliases.
--   Here: each employee matched to their manager (also in
--   employees). Ananya has no manager → manager name is NULL.
-----
-- Self join – employee to manager
SELECT
    e.empName AS employee,
    m.empName AS manager
FROM employees AS e
LEFT JOIN employees AS m ON e.managerID = m.empID
ORDER BY e.empID;

```

6. Output

Each SELECT from the script, executed in SQLite. Unmatched columns print as NULL (highlighted). CREATE and INSERT output is omitted here; the statements are in section 5.

6.1 Base table — departments

```
SQL> SELECT * FROM departments;
```

deptID	deptName
1	School of Engineering and Technology
2	School of Management
3	School of Law
4	School of Agricultural Sciences
5	School of Medical Sciences
6	School of Humanities

6.2 Base table — employees

```
SQL> SELECT * FROM employees;
```

empID	empName	deptID	managerID	salary
1	Ananya Sharma	1	NULL	92000
2	Rohan Mehta	2	1	78000
3	Priya Nair	3	1	85000
4	Vikram Singh	4	2	76000
5	Fatima Khan	1	1	88000
6	Arjun Patel	5	3	95000
7	Neha Gupta	99	2	54000

6.3 CROSS JOIN — row count

```
SQL> SELECT COUNT(*) AS cross_join_row_count
FROM departments
CROSS JOIN employees;
```

cross_join_row_count
42

6.4 CROSS JOIN — first 8 pairs

```
SQL> SELECT d.deptName, e.empName
      FROM departments AS d
      CROSS JOIN employees AS e
      ORDER BY d.deptID, e.empID
      LIMIT 8;
```

deptName	empName
School of Engineering and Technology	Ananya Sharma
School of Engineering and Technology	Rohan Mehta
School of Engineering and Technology	Priya Nair
School of Engineering and Technology	Vikram Singh
School of Engineering and Technology	Fatima Khan
School of Engineering and Technology	Arjun Patel
School of Engineering and Technology	Neha Gupta
School of Management	Ananya Sharma

6.5 NATURAL JOIN

```
SQL> SELECT empID, empName, deptName, salary
      FROM employees
      NATURAL JOIN departments;
```

empID	empName	deptName	salary
1	Ananya Sharma	School of Engineering and Technology	92000
2	Rohan Mehta	School of Management	78000
3	Priya Nair	School of Law	85000
4	Vikram Singh	School of Agricultural Sciences	76000
5	Fatima Khan	School of Engineering and Technology	88000
6	Arjun Patel	School of Medical Sciences	95000

6.6 INNER JOIN

```
SQL> SELECT e.empID, e.empName, d.deptName, e.salary
      FROM employees AS e
      INNER JOIN departments AS d ON e.deptID = d.deptID;
```

empID	empName	deptName	salary
1	Ananya Sharma	School of Engineering and Technology	92000
2	Rohan Mehta	School of Management	78000
3	Priya Nair	School of Law	85000
4	Vikram Singh	School of Agricultural Sciences	76000
5	Fatima Khan	School of Engineering and Technology	88000
6	Arjun Patel	School of Medical Sciences	95000

6.7 LEFT OUTER JOIN

```
SQL> SELECT e.empID, e.empName, e.deptID, d.deptName, e.salary
FROM employees AS e
LEFT OUTER JOIN departments AS d ON e.deptID = d.deptID;
```

empID	empName	deptID	deptName	salary
1	Ananya Sharma	1	School of Engineering and Technology	92000
2	Rohan Mehta	2	School of Management	78000
3	Priya Nair	3	School of Law	85000
4	Vikram Singh	4	School of Agricultural Sciences	76000
5	Fatima Khan	1	School of Engineering and Technology	88000
6	Arjun Patel	5	School of Medical Sciences	95000
7	Neha Gupta	99	NULL	54000

6.8 RIGHT OUTER JOIN

```
SQL> SELECT e.empID, e.empName, d.deptID, d.deptName, e.salary
FROM employees AS e
RIGHT OUTER JOIN departments AS d ON e.deptID = d.deptID;
```

empID	empName	deptID	deptName	salary
1	Ananya Sharma	1	School of Engineering and Technology	92000
2	Rohan Mehta	2	School of Management	78000
3	Priya Nair	3	School of Law	85000
4	Vikram Singh	4	School of Agricultural Sciences	76000
5	Fatima Khan	1	School of Engineering and Technology	88000
6	Arjun Patel	5	School of Medical Sciences	95000
NULL	NULL	6	School of Humanities	NULL

6.9 SELF JOIN — employee and manager

```
SQL> SELECT
    e.empName AS employee,
    m.empName AS manager
FROM employees AS e
LEFT JOIN employees AS m ON e.managerID = m.empID
ORDER BY e.empID;
```

employee	manager
Ananya Sharma	NULL
Rohan Mehta	Ananya Sharma
Priya Nair	Ananya Sharma
Vikram Singh	Rohan Mehta
Fatima Khan	Ananya Sharma
Arjun Patel	Priya Nair
Neha Gupta	Rohan Mehta

7. Results

Row counts returned by each join on this dataset:

Join type	Rows	Unmatched handling
CROSS JOIN	42	No condition — every dept × every employee
NATURAL JOIN	6	Drops non-matching rows from both sides
INNER JOIN	6	Same six rows as NATURAL JOIN here
LEFT OUTER JOIN	7	Keeps Neha Gupta (deptID 99); deptName is NULL
RIGHT OUTER JOIN	7	Keeps Humanities (deptID 6); employee cols are NULL
SELF JOIN	7	All employees; Ananya has NULL manager (top of chain)

CROSS JOIN produces the largest result because it ignores relationships entirely. INNER JOIN and NATURAL JOIN return only the six employees whose deptID exists in departments. Outer joins add the orphan employee (left) or orphan department (right). SELF JOIN does not change row count here because a LEFT JOIN keeps every employee and simply leaves the manager column null for Ananya Sharma, who has no manager.

8. Conclusion

SQL joins answer different questions about related data. CROSS JOIN asks “every combination?” INNER JOIN asks “only matches?” Outer joins ask “keep everyone from this side even without a match?” SELF JOIN asks “how do rows in this table relate to other rows in the same table?” Choosing the right join depends on whether you need all combinations, only matches, or orphan rows from a specific side.

Ex – 4

SELECT Queries

1. Aim

To answer the lab questions below: create an employee table with the given attributes, insert 10 rows, and run the listed SELECT queries (DISTINCT, WHERE, BETWEEN, IN, ORDER BY). Then create compatible cse and mechanical student tables and combine them with UNION, UNION ALL, and INTERSECT.

2. Theory

2.1 SELECT

SELECT reads rows from a relation. SELECT * returns every column; naming columns returns only those attributes. A query without a WHERE clause returns every row that currently exists in the table.

2.2 DISTINCT

SELECT DISTINCT removes duplicate values from the result. Question 3 on the lab sheet asks for a “distant” department number; that is DISTINCT. SELECT department_no lists a department once per employee (question 4); SELECT DISTINCT department_no lists each department number only once.

2.3 WHERE

The WHERE clause keeps rows that satisfy a condition. Comparison operators such as > and <> filter numbers and strings. OR keeps a row if either predicate is true (city is Delhi or Mumbai). City values in this lab are stored in title case and matched with the same casing so the filters work in SQLite without extra case-conversion functions.

2.4 BETWEEN

BETWEEN low AND high is an inclusive range: the endpoints are part of the result. NOT BETWEEN is the complement — salaries strictly below 30000 or strictly above 50000. It is equivalent to salary < 30000 OR salary > 50000 when the range is 30000 to 50000.

2.5 IN

IN (...) tests membership in a list. city IN ('Delhi', 'Mumbai', 'Chennai', 'Bangalore') keeps those four cities. NOT IN keeps every city that is not in the list — here Bangalore, Hyderabad, and Pune remain when Mumbai, Delhi, and Chennai are excluded.

2.6 ORDER BY

ORDER BY sorts the result. ASC (the default) lists names A→Z; DESC lists them Z→A. Sorting does not change stored rows; it only changes the order of the result set.

2.7 UNION

UNION stacks two compatible SELECT results and keeps each distinct row once. Both sides must return the same number of columns with comparable types. Duplicate rows that appear in both cse and mechanical are listed only once. Column names in the result come from the first SELECT.

2.8 UNION ALL

UNION ALL also stacks two results but does **not** remove duplicates. A row that exists in both tables appears twice. Use it when every occurrence matters, or when the extra distinct pass of UNION is unnecessary. Here five CSE rows plus five Mechanical rows yield ten rows.

2.9 INTERSECT

INTERSECT keeps rows that appear in **both** results. Entire rows must match (roll_no, student_name, and city). Amit Verma / Delhi and Sneha Reddy / Hyderabad were inserted into both tables, so the intersection has two rows. SQLite and SQL Server both support INTERSECT.

3. Schema

Three tables. employee holds eight attributes for questions 1–13. sr_no is the employee number (primary key). manager stores the manager's name, or NULL for the top manager. cse and mechanical are compatible three-column student lists for questions 14–22.

Attribute	Type	Constraint	Role
sr_no	INTEGER	PRIMARY KEY	Employee number
employee_name	TEXT	NOT NULL	Employee name
job	TEXT	NOT NULL	Job title
manager	TEXT	NULL allowed	Manager name
hire_date	DATE	NOT NULL	Date of joining
salary	NUMERIC(10,2)	NOT NULL, CHECK > 0	Monthly salary
department_no	INTEGER	NOT NULL	Department number
city	TEXT	NOT NULL	Work city

cse and mechanical share the same three attributes so they can be combined with set operators:

Attribute	Type	Constraint	Role
roll_no	INTEGER	PRIMARY KEY	Student roll number
student_name	TEXT	NOT NULL	Student name
city	TEXT	NOT NULL	Home city

4. Questions

The lab sheet numbered both CREATE and INSERT as **1**; they are shown here as **1(a)** and **1(b)**. Question 3 wrote “distant”; the SQL keyword is DISTINCT. Employee number is stored as sr_no. Questions **14–22** are the CSE / Mechanical UNION, UNION ALL, and INTERSECT exercise: create both student tables, list their rows, then combine them.

1(a). Create a table employee with the following attributes:

- sr no
- employee name
- job
- manager
- hire date
- salary
- department no
- city

- 1(b). Insert 10 rows in the above mentioned table.
2. SELECT all the details from the employee table.
3. Select only DISTINCT department number from the employee table.
The lab sheet wrote "distant" for this question.
4. SELECT all department numbers from the employee table.
5. SELECT employee name, salary WHERE salary is > 30000 rs from the employee table.
6. SELECT employee name, hire date WHERE city is not Delhi from the employee table.
7. SELECT employee name, hire date WHERE city is either Delhi or Mumbai from the employee table.
8. SELECT employee number, employee name WHERE salary is BETWEEN 30000 rs and 50000 rs.
9. SELECT employee number, employee name WHERE salary is NOT BETWEEN 30000 and 50000 from the employee table.
10. SELECT all the details WHERE city is among the following: Delhi, Mumbai, Chennai, Bangalore.
11. SELECT all the details WHERE city is not Mumbai, Delhi or Chennai from the employee table.
12. SELECT all the details from the employee table and ORDER BY employee names in descending order.
13. Display the list of employees in ascending order from the employee table.
14. Create a table cse (CSE students) with the following attributes:
 - roll no
 - student name
 - city
15. Insert 5 rows in the cse table.
16. Create a table mechanical (Mechanical students) with the following attributes:
 - roll no
 - student name
 - city
17. Insert 5 rows in the mechanical table.
Two rows also appear in cse (Amit Verma / Delhi and Sneha Reddy / Hyderabad) so INTERSECT is non-empty.
18. SELECT all the details from the cse table.
19. SELECT all the details from the mechanical table.
20. SELECT the UNION of cse and mechanical.
21. SELECT the UNION ALL of cse and mechanical.
22. SELECT the INTERSECT of cse and mechanical.

5. Procedure

Drop employee, cse, and mechanical if they already exist so the script can be re-run. Then answer each lab question in order. City values on employee are stored in title case (Delhi, Mumbai, ...) and matched with that

same casing. Set-operation results are ordered by roll_no. The SQL for each question:

Q. No.	SQL solution
1(a)	<pre>CREATE TABLE employee (sr_no INTEGER PRIMARY KEY, employee_name TEXT NOT NULL, job TEXT NOT NULL, manager TEXT, hire_date DATE NOT NULL, salary NUMERIC(10, 2) NOT NULL CHECK (salary > 0), department_no INTEGER NOT NULL, city TEXT NOT NULL);</pre>
1(b)	<pre>INSERT INTO employee (sr_no, employee_name, job, manager, hire_date, salary, department_no, city) VALUES (1, 'Rajesh Kumar', 'General Manager', NULL, '2018-03-12', 85000.00, 10, 'Delhi'), (2, 'Ananya Sharma', 'Software Engineer', 'Rajesh Kumar', '2021-07-01', 45000.00, 20, 'Bangalore'), (3, 'Rohan Mehta', 'Accountant', 'Rajesh Kumar', '2020-01-15', 28000.00, 30, 'Mumbai'), (4, 'Priya Nair', 'HR Executive', 'Rajesh Kumar', '2022-04-20', 32000.00, 40, 'Chennai'), (5, 'Vikram Singh', 'Sales Manager', 'Rajesh Kumar', '2019-11-08', 52000.00, 50, 'Hyderabad'), (6, 'Fatima Khan', 'Software Engineer', 'Ananya Sharma', '2023-02-14', 38000.00, 20, 'Bangalore'), (7, 'Arjun Patel', 'Clerk', 'Rohan Mehta', '2024-06-01', 18000.00, 30, 'Pune'), (8, 'Neha Gupta', 'Marketing Analyst', 'Rajesh Kumar', '2021-09-10', 35000.00, 60, 'Delhi'), (9, 'Karan Iyer', 'Database Admin', 'Ananya Sharma', '2020-12-05', 48000.00, 20, 'Mumbai'), (10, 'Meera Joshi', 'Receptionist', 'Priya Nair', '2023-08-22', 22000.00, 40, 'Chennai');</pre>
2	<pre>SELECT * FROM employee;</pre>
3	<pre>SELECT DISTINCT department_no FROM employee;</pre>
4	<pre>SELECT department_no FROM employee;</pre>
5	<pre>SELECT employee_name, salary FROM employee WHERE salary > 30000;</pre>
6	<pre>SELECT employee_name, hire_date FROM employee WHERE city <> 'Delhi';</pre>
7	<pre>SELECT employee_name, hire_date FROM employee WHERE city = 'Delhi' OR city = 'Mumbai';</pre>
8	<pre>SELECT sr_no, employee_name FROM employee WHERE salary BETWEEN 30000 AND 50000;</pre>
9	<pre>SELECT sr_no, employee_name FROM employee WHERE salary NOT BETWEEN 30000 AND 50000;</pre>
10	<pre>SELECT * FROM employee WHERE city IN ('Delhi', 'Mumbai', 'Chennai', 'Bangalore');</pre>
11	<pre>SELECT * FROM employee WHERE city NOT IN ('Mumbai', 'Delhi', 'Chennai');</pre>
12	<pre>SELECT * FROM employee ORDER BY employee_name DESC;</pre>
13	<pre>SELECT * FROM employee ORDER BY employee_name ASC;</pre>
14	<pre>CREATE TABLE cse (roll_no INTEGER PRIMARY KEY, student_name TEXT NOT NULL, city TEXT NOT NULL);</pre>

Q. No.	SQL solution
15	<pre> INSERT INTO cse (roll_no, student_name, city) VALUES (1, 'Amit Verma', 'Delhi'), (2, 'Rahul Das', 'Kolkata'), (3, 'Sneha Reddy', 'Hyderabad'), (4, 'Isha Kapoor', 'Chandigarh'), (5, 'Dev Patel', 'Ahmedabad'); </pre>
16	<pre> CREATE TABLE mechanical (roll_no INTEGER PRIMARY KEY, student_name TEXT NOT NULL, city TEXT NOT NULL); </pre>
17	<pre> INSERT INTO mechanical (roll_no, student_name, city) VALUES (1, 'Amit Verma', 'Delhi'), (3, 'Sneha Reddy', 'Hyderabad'), (6, 'Mohit Jain', 'Jaipur'), (7, 'Kavya Menon', 'Kochi'), (8, 'Tushar Rao', 'Nagpur'); </pre>
18	<pre> SELECT * FROM cse; </pre>
19	<pre> SELECT * FROM mechanical; </pre>
20	<pre> SELECT * FROM cse UNION SELECT * FROM mechanical ORDER BY roll_no; </pre>
21	<pre> SELECT * FROM cse UNION ALL SELECT * FROM mechanical ORDER BY roll_no; </pre>
22	<pre> SELECT * FROM cse INTERSECT SELECT * FROM mechanical ORDER BY roll_no; </pre>

6. Source Code

SQLite script used in the lab compiler (09-09-2026/employee.sql). For the class SQL Server, run employee.sqlserver.sql.

```

-- =====
-- DBMS Lab · 09-09-2026
-- Employee table: CREATE, seed 10 rows, then SELECT queries
-- DISTINCT, WHERE, BETWEEN, IN, ORDER BY
-- Then CSE / Mechanical student tables: UNION, UNION ALL, INTERSECT
-- SQLite (sql.js compiler / DB Browser / sqlite3)
-- For the class SQL Server, run employee.sqlserver.sql instead.
--
-- Lab questions (1(a)/1(b) = sheet's two "1." items; "distant" = DISTINCT)
-- 1(a). Create a table employee with the following attributes:
--     sr no, employee name, job, manager, hire date, salary,
--     department no, city
-- 1(b). Insert 10 rows in the above mentioned table
-- 2. SELECT all the details from the employee table
-- 3. Select only DISTINCT department number from employee table
--     (lab sheet wrote "distant")
-- 4. SELECT all department numbers from the employee table
-- 5. SELECT employee name, salary WHERE salary is > 30000 rs
--     from the employee table
-- 6. SELECT employee name, hire date WHERE city is not Delhi
--     from the employee table
-- 7. SELECT employee name, hire date WHERE city is either
--     Delhi or Mumbai from the employee table
-- 8. SELECT employee number, employee name WHERE salary is
--     BETWEEN 30000 rs and 50000 rs
-- 9. SELECT employee number, employee name WHERE salary is
--     NOT BETWEEN 30000 and 50000 from the employee table
-- 10. SELECT all the details WHERE city is among the following:
--     Delhi, Mumbai, Chennai, Bangalore
-- 11. SELECT all the details WHERE city is not Mumbai, Delhi
--     or Chennai from the employee table
-- 12. SELECT all the details from the employee table and
--     ORDER BY employee names in descending order
-- 13. Display the list of employees in ascending order from
--     the employee table.

```

```

-- 14. Create a table cse (CSE students) with roll no, student name, city
-- 15. Insert 5 rows in the cse table
-- 16. Create a table mechanical (Mechanical students) with the same attributes
-- 17. Insert 5 rows in the mechanical table
--      (two rows also appear in cse so INTERSECT is non-empty)
-- 18. SELECT all the details from the cse table
-- 19. SELECT all the details from the mechanical table
-- 20. SELECT the UNION of cse and mechanical
-- 21. SELECT the UNION ALL of cse and mechanical
-- 22. SELECT the INTERSECT of cse and mechanical
-- =====

DROP TABLE IF EXISTS employee;

-- =====
-- 1(a). Create a table employee with the following attributes:
--      sr no, employee name, job, manager, hire date, salary,
--      department no, city
-- =====
CREATE TABLE employee (
    sr_no          INTEGER PRIMARY KEY,
    employee_name  TEXT NOT NULL,
    job            TEXT NOT NULL,
    manager        TEXT,
    hire_date      DATE NOT NULL,
    salary         NUMERIC(10, 2) NOT NULL CHECK (salary > 0),
    department_no  INTEGER NOT NULL,
    city          TEXT NOT NULL
);

-- =====
-- 1(b). Insert 10 rows in the above mentioned table
--      • salaries below, inside, and above 30000-50000
--      • cities: Delhi, Mumbai, Chennai, Bangalore, Hyderabad, Pune
--      • mixed department numbers (some duplicates)
--      • Rajesh Kumar is the top manager (manager IS NULL)
-- =====
INSERT INTO employee
(sr_no, employee_name, job, manager, hire_date, salary, department_no, city)
VALUES
(1, 'Rajesh Kumar', 'General Manager', NULL, '2018-03-12', 85000.00, 10, 'Delhi'),
(2, 'Ananya Sharma', 'Software Engineer', 'Rajesh Kumar', '2021-07-01', 45000.00, 20, 'Bangalore'),
(3, 'Rohan Mehta', 'Accountant', 'Rajesh Kumar', '2020-01-15', 28000.00, 30, 'Mumbai'),
(4, 'Priya Nair', 'HR Executive', 'Rajesh Kumar', '2022-04-20', 32000.00, 40, 'Chennai'),
(5, 'Vikram Singh', 'Sales Manager', 'Rajesh Kumar', '2019-11-08', 52000.00, 50, 'Hyderabad'),
(6, 'Fatima Khan', 'Software Engineer', 'Ananya Sharma', '2023-02-14', 38000.00, 20, 'Bangalore'),
(7, 'Arjun Patel', 'Clerk', 'Rohan Mehta', '2024-06-01', 18000.00, 30, 'Pune'),
(8, 'Neha Gupta', 'Marketing Analyst', 'Rajesh Kumar', '2021-09-10', 35000.00, 60, 'Delhi'),
(9, 'Karan Iyer', 'Database Admin', 'Ananya Sharma', '2020-12-05', 48000.00, 20, 'Mumbai'),
(10, 'Meera Joshi', 'Receptionist', 'Priya Nair', '2023-08-22', 22000.00, 40, 'Chennai');

-- =====
-- 2. select all the details from the employee table
-- =====
-- All the details from employee
SELECT * FROM employee;

-- =====
-- 3. Select only DISTINCT department number from employee table
--      (lab sheet wrote "distant")
-- =====
-- Distant (DISTINCT) department number
SELECT DISTINCT department_no FROM employee;

-- =====
-- 4. select all department numbers from employee table
-- =====
-- All department numbers
SELECT department_no FROM employee;

-- =====
-- 5. select employee name, salary where salary is >30000 rs
--      from employee table
-- =====
-- Salary > 30000 rs
SELECT employee_name, salary
FROM employee
WHERE salary > 30000;

-- =====
-- 6. select employee name, hire date where city is not delhi
--      from employee table
-- =====
-- City is not delhi
SELECT employee_name, hire_date
FROM employee
WHERE city <> 'Delhi';

-- =====
-- 7. select employee name, hire date where city is either

```

```

-- delhi or mumbai from employee table
-----
-- City is either delhi or mumbai
SELECT employee_name, hire_date
FROM employee
WHERE city = 'Delhi' OR city = 'Mumbai';

-----
-- 8. select employee number, employee name where salary is
--    between 30000 rs and 50000 rs
-----
-- Salary between 30000 rs and 50000 rs
SELECT sr_no, employee_name
FROM employee
WHERE salary BETWEEN 30000 AND 50000;

-----
-- 9. select employee number, employee name where salary is
--    not between 30000 and 50000 from employee table
-----
-- Salary not between 30000 and 50000
SELECT sr_no, employee_name
FROM employee
WHERE salary NOT BETWEEN 30000 AND 50000;

-----
-- 10. select all the details where city is among the following:
--     delhi, mumbai, chennai, Bangalore
-----
-- City among delhi, mumbai, chennai, Bangalore
SELECT *
FROM employee
WHERE city IN ('Delhi', 'Mumbai', 'Chennai', 'Bangalore');

-----
-- 11. select all the details where city is not mumbai, delhi
--     or chennai from employee table
-----
-- City is not mumbai, delhi or chennai
SELECT *
FROM employee
WHERE city NOT IN ('Mumbai', 'Delhi', 'Chennai');

-----
-- 12. select all the details from the employee table and
--     order by employee names in descending order.
-----
-- Order by employee names descending
SELECT *
FROM employee
ORDER BY employee_name DESC;

-----
-- 13. display the list of employees in ascending order from
--     employee table.
-----
-- Display employees in ascending order
SELECT *
FROM employee
ORDER BY employee_name ASC;

-----
-- =====
-- CSE / Mechanical student tables – UNION, UNION ALL, INTERSECT
-- Compatible columns (roll_no, student_name, city). Two identical
-- rows appear in both tables so INTERSECT returns a non-empty set.
-- =====

DROP TABLE IF EXISTS cse;
DROP TABLE IF EXISTS mechanical;

-----
-- 14. Create a table cse (CSE students) with the following
--     attributes: roll no, student name, city
-----
CREATE TABLE cse (
    roll_no      INTEGER PRIMARY KEY,
    student_name TEXT NOT NULL,
    city         TEXT NOT NULL
);

-----
-- 15. Insert 5 rows in the cse table
-----
INSERT INTO cse (roll_no, student_name, city)
VALUES
    (1, 'Amit Verma', 'Delhi'),
    (2, 'Rahul Das', 'Kolkata'),
    (3, 'Sneha Reddy', 'Hyderabad'),
    (4, 'Isha Kapoor', 'Chandigarh'),
    (5, 'Dev Patel', 'Ahmedabad');

```

```

-----
-- 16. Create a table mechanical (Mechanical students) with the
--      following attributes: roll no, student name, city
-----
CREATE TABLE mechanical (
    roll_no      INTEGER PRIMARY KEY,
    student_name TEXT NOT NULL,
    city         TEXT NOT NULL
);

-----
-- 17. Insert 5 rows in the mechanical table
--      Two rows match cse (Amit Verma / Delhi, Sneha Reddy / Hyderabad)
-----
INSERT INTO mechanical (roll_no, student_name, city)
VALUES
    (1, 'Amit Verma', 'Delhi'),
    (3, 'Sneha Reddy', 'Hyderabad'),
    (6, 'Mohit Jain', 'Jaipur'),
    (7, 'Kavya Menon', 'Kochi'),
    (8, 'Tushar Rao', 'Nagpur');

-----
-- 18. SELECT all the details from the cse table
-----
-- All the details from CSE
SELECT * FROM cse;

-----
-- 19. SELECT all the details from the mechanical table
-----
-- All the details from Mechanical
SELECT * FROM mechanical;

-----
-- 20. SELECT the UNION of cse and mechanical
--      (unique rows from either table)
-----
-- UNION of CSE and Mechanical
SELECT * FROM cse
UNION
SELECT * FROM mechanical
ORDER BY roll_no;

-----
-- 21. SELECT the UNION ALL of cse and mechanical
--      (all rows, including duplicates)
-----
-- UNION ALL of CSE and Mechanical
SELECT * FROM cse
UNION ALL
SELECT * FROM mechanical
ORDER BY roll_no;

-----
-- 22. SELECT the INTERSECT of cse and mechanical
--      (rows that appear in both tables)
-----
-- INTERSECT of CSE and Mechanical
SELECT * FROM cse
INTERSECT
SELECT * FROM mechanical
ORDER BY roll_no;

```

7. Output

The script was executed in SQLite in memory. CREATE and INSERT print a status line; each SELECT prints its result set. NULL in the manager column is shown as an empty cell.

```

SQL> DROP TABLE IF EXISTS employee;
-- OK

SQL> CREATE TABLE employee (
    sr_no      INTEGER PRIMARY KEY,
    employee_name TEXT NOT NULL,
    job        TEXT NOT NULL,
    manager     TEXT,
    hire_date   DATE NOT NULL,
    salary      NUMERIC(10, 2) NOT NULL CHECK (salary > 0),
    department_no INTEGER NOT NULL,
    city       TEXT NOT NULL
);
-- OK

SQL> INSERT INTO employee
    (sr_no, employee_name, job, manager, hire_date, salary, department_no, city)
VALUES
    (1, 'Rajesh Kumar', 'General Manager', NULL, '2018-03-12', 85000.00,
    10, 'Delhi'),

```

```

        (2, 'Ananya Sharma', 'Software Engineer', 'Rajesh Kumar', '2021-07-01', 45000.00,
20, 'Bangalore'),
        (3, 'Rohan Mehta', 'Accountant', 'Rajesh Kumar', '2020-01-15', 28000.00,
30, 'Mumbai'),
        (4, 'Priya Nair', 'HR Executive', 'Rajesh Kumar', '2022-04-20', 32000.00,
40, 'Chennai'),
        (5, 'Vikram Singh', 'Sales Manager', 'Rajesh Kumar', '2019-11-08', 52000.00,
50, 'Hyderabad'),
        (6, 'Fatima Khan', 'Software Engineer', 'Ananya Sharma', '2023-02-14', 38000.00,
20, 'Bangalore'),
        (7, 'Arjun Patel', 'Clerk', 'Rohan Mehta', '2024-06-01', 18000.00,
30, 'Pune'),
        (8, 'Neha Gupta', 'Marketing Analyst', 'Rajesh Kumar', '2021-09-10', 35000.00,
60, 'Delhi'),
        (9, 'Karan Iyer', 'Database Admin', 'Ananya Sharma', '2020-12-05', 48000.00,
20, 'Mumbai'),
        (10, 'Meera Joshi', 'Receptionist', 'Priya Nair', '2023-08-22', 22000.00,
40, 'Chennai');
-- 10 row(s) inserted

```

```

SQL> SELECT * FROM employee;
sr_no  employee_name  job              manager      hire_date  salary  department_no  city
-----
1      Rajesh Kumar   General Manager
2      Ananya Sharma  Software Engineer Rajesh Kumar  2021-07-01 45000   20             Bangalore
3      Rohan Mehta    Accountant       Rajesh Kumar  2020-01-15 28000   30             Mumbai
4      Priya Nair     HR Executive     Rajesh Kumar  2022-04-20 32000   40             Chennai
5      Vikram Singh   Sales Manager    Rajesh Kumar  2019-11-08 52000   50             Hyderabad
6      Fatima Khan    Software Engineer Ananya Sharma 2023-02-14 38000   20             Bangalore
7      Arjun Patel    Clerk           Rohan Mehta   2024-06-01 18000   30             Pune
8      Neha Gupta     Marketing Analyst Rajesh Kumar  2021-09-10 35000   60             Delhi
9      Karan Iyer     Database Admin   Ananya Sharma 2020-12-05 48000   20             Mumbai
10     Meera Joshi    Receptionist     Priya Nair    2023-08-22 22000   40             Chennai

```

```

SQL> SELECT DISTINCT department_no FROM employee;
department_no
-----
10
20
30
40
50
60

```

```

SQL> SELECT department_no FROM employee;
department_no
-----
10
20
30
40
50
20
30
60
20
40

```

```

SQL> SELECT employee_name, salary
FROM employee
WHERE salary > 30000;
employee_name  salary
-----
Rajesh Kumar   45000
Ananya Sharma  45000
Priya Nair     32000
Vikram Singh   52000
Fatima Khan    38000
Neha Gupta     35000
Karan Iyer     48000

```

```

SQL> SELECT employee_name, hire_date
FROM employee
WHERE city <> 'Delhi';
employee_name  hire_date
-----
Ananya Sharma  2021-07-01
Rohan Mehta    2020-01-15
Priya Nair     2022-04-20
Vikram Singh   2019-11-08
Fatima Khan    2023-02-14
Arjun Patel    2024-06-01
Karan Iyer     2020-12-05
Meera Joshi    2023-08-22

```

```

SQL> SELECT employee_name, hire_date
FROM employee
WHERE city = 'Delhi' OR city = 'Mumbai';
employee_name  hire_date
-----
Rajesh Kumar   2018-03-12
Rohan Mehta    2020-01-15
Neha Gupta     2021-09-10
Karan Iyer     2020-12-05

```

```

SQL> SELECT sr_no, employee_name
FROM employee
WHERE salary BETWEEN 30000 AND 50000;
sr_no  employee_name
-----
2      Ananya Sharma
4      Priya Nair

```

```
6      Fatima Khan
8      Neha Gupta
9      Karan Iyer
```

```
SQL> SELECT sr_no, employee_name
      FROM employee
      WHERE salary NOT BETWEEN 30000 AND 50000;
sr_no  employee_name
-----
1      Rajesh Kumar
3      Rohan Mehta
5      Vikram Singh
7      Arjun Patel
10     Meera Joshi
```

```
SQL> SELECT *
      FROM employee
      WHERE city IN ('Delhi', 'Mumbai', 'Chennai', 'Bangalore');
sr_no  employee_name  job            manager  hire_date  salary  department_no  city
-----
1      Rajesh Kumar  General Manager  manager  2018-03-12  85000   10             Delhi
2      Ananya Sharma  Software Engineer  Rajesh Kumar  2021-07-01  45000   20             Bangalore
3      Rohan Mehta  Accountant      Rajesh Kumar  2020-01-15  28000   30             Mumbai
4      Priya Nair   HR Executive    Rajesh Kumar  2022-04-20  32000   40             Chennai
6      Fatima Khan  Software Engineer  Ananya Sharma  2023-02-14  38000   20             Bangalore
8      Neha Gupta  Marketing Analyst Rajesh Kumar  2021-09-10  35000   60             Delhi
9      Karan Iyer  Database Admin  Ananya Sharma  2020-12-05  48000   20             Mumbai
10     Meera Joshi  Receptionist    Priya Nair    2023-08-22  22000   40             Chennai
```

```
SQL> SELECT *
      FROM employee
      WHERE city NOT IN ('Mumbai', 'Delhi', 'Chennai');
sr_no  employee_name  job            manager  hire_date  salary  department_no  city
-----
2      Ananya Sharma  Software Engineer  Rajesh Kumar  2021-07-01  45000   20             Bangalore
5      Vikram Singh  Sales Manager    Rajesh Kumar  2019-11-08  52000   50             Hyderabad
6      Fatima Khan  Software Engineer  Ananya Sharma  2023-02-14  38000   20             Bangalore
7      Arjun Patel  Clerk           Rohan Mehta  2024-06-01  18000   30             Pune
```

```
SQL> SELECT *
      FROM employee
      ORDER BY employee_name DESC;
sr_no  employee_name  job            manager  hire_date  salary  department_no  city
-----
5      Vikram Singh  Sales Manager    Rajesh Kumar  2019-11-08  52000   50             Hyderabad
3      Rohan Mehta  Accountant      Rajesh Kumar  2020-01-15  28000   30             Mumbai
1      Rajesh Kumar  General Manager  Rajesh Kumar  2018-03-12  85000   10             Delhi
4      Priya Nair   HR Executive    Rajesh Kumar  2022-04-20  32000   40             Chennai
8      Neha Gupta  Marketing Analyst Rajesh Kumar  2021-09-10  35000   60             Delhi
10     Meera Joshi  Receptionist    Priya Nair    2023-08-22  22000   40             Chennai
9      Karan Iyer  Database Admin  Ananya Sharma  2020-12-05  48000   20             Mumbai
6      Fatima Khan  Software Engineer  Ananya Sharma  2023-02-14  38000   20             Bangalore
7      Arjun Patel  Clerk           Rohan Mehta  2024-06-01  18000   30             Pune
2      Ananya Sharma  Software Engineer  Rajesh Kumar  2021-07-01  45000   20             Bangalore
```

```
SQL> SELECT *
      FROM employee
      ORDER BY employee_name ASC;
sr_no  employee_name  job            manager  hire_date  salary  department_no  city
-----
2      Ananya Sharma  Software Engineer  Rajesh Kumar  2021-07-01  45000   20             Bangalore
7      Arjun Patel  Clerk           Rohan Mehta  2024-06-01  18000   30             Pune
6      Fatima Khan  Software Engineer  Ananya Sharma  2023-02-14  38000   20             Bangalore
9      Karan Iyer  Database Admin  Ananya Sharma  2020-12-05  48000   20             Mumbai
10     Meera Joshi  Receptionist    Priya Nair    2023-08-22  22000   40             Chennai
8      Neha Gupta  Marketing Analyst Rajesh Kumar  2021-09-10  35000   60             Delhi
4      Priya Nair   HR Executive    Rajesh Kumar  2022-04-20  32000   40             Chennai
1      Rajesh Kumar  General Manager  Rajesh Kumar  2018-03-12  85000   10             Delhi
3      Rohan Mehta  Accountant      Rajesh Kumar  2020-01-15  28000   30             Mumbai
5      Vikram Singh  Sales Manager    Rajesh Kumar  2019-11-08  52000   50             Hyderabad
```

```
SQL> DROP TABLE IF EXISTS cse;
-- OK
```

```
SQL> DROP TABLE IF EXISTS mechanical;
-- OK
```

```
SQL> CREATE TABLE cse (
      roll_no      INTEGER PRIMARY KEY,
      student_name  TEXT NOT NULL,
      city          TEXT NOT NULL
);
-- OK
```

```
SQL> INSERT INTO cse (roll_no, student_name, city)
      VALUES
      (1, 'Amit Verma', 'Delhi'),
      (2, 'Rahul Das', 'Kolkata'),
      (3, 'Sneha Reddy', 'Hyderabad'),
      (4, 'Isha Kapoor', 'Chandigarh'),
      (5, 'Dev Patel', 'Ahmedabad');
-- 5 row(s) inserted
```

```
SQL> CREATE TABLE mechanical (
      roll_no      INTEGER PRIMARY KEY,
      student_name  TEXT NOT NULL,
      city          TEXT NOT NULL
);
-- OK
```

```
SQL> INSERT INTO mechanical (roll_no, student_name, city)
      VALUES
```

```

        (1, 'Amit Verma', 'Delhi'),
        (3, 'Sneha Reddy', 'Hyderabad'),
        (6, 'Mohit Jain', 'Jaipur'),
        (7, 'Kavya Menon', 'Kochi'),
        (8, 'Tushar Rao', 'Nagpur');
-- 5 row(s) inserted

```

```

SQL> SELECT * FROM cse;
roll_no  student_name  city
-----
1        Amit Verma  Delhi
2        Rahul Das   Kolkata
3        Sneha Reddy  Hyderabad
4        Isha Kapoor  Chandigarh
5        Dev Patel    Ahmedabad

```

```

SQL> SELECT * FROM mechanical;
roll_no  student_name  city
-----
1        Amit Verma  Delhi
3        Sneha Reddy  Hyderabad
6        Mohit Jain   Jaipur
7        Kavya Menon  Kochi
8        Tushar Rao   Nagpur

```

```

SQL> SELECT * FROM cse
      UNION
      SELECT * FROM mechanical
      ORDER BY roll_no;
roll_no  student_name  city
-----
1        Amit Verma  Delhi
2        Rahul Das   Kolkata
3        Sneha Reddy  Hyderabad
4        Isha Kapoor  Chandigarh
5        Dev Patel    Ahmedabad
6        Mohit Jain   Jaipur
7        Kavya Menon  Kochi
8        Tushar Rao   Nagpur

```

```

SQL> SELECT * FROM cse
      UNION ALL
      SELECT * FROM mechanical
      ORDER BY roll_no;
roll_no  student_name  city
-----
1        Amit Verma  Delhi
1        Amit Verma  Delhi
2        Rahul Das   Kolkata
3        Sneha Reddy  Hyderabad
3        Sneha Reddy  Hyderabad
4        Isha Kapoor  Chandigarh
5        Dev Patel    Ahmedabad
6        Mohit Jain   Jaipur
7        Kavya Menon  Kochi
8        Tushar Rao   Nagpur

```

```

SQL> SELECT * FROM cse
      INTERSECT
      SELECT * FROM mechanical
      ORDER BY roll_no;
roll_no  student_name  city
-----
1        Amit Verma  Delhi
3        Sneha Reddy  Hyderabad

```

8. Results

Ten employees load successfully, then five CSE students and five Mechanical students. Each lab question returns a non-empty result on this seed data:

Q. No.	Rows	What the result shows
1(a)	—	Table employee created with 8 attributes
1(b)	10	10 rows inserted
2	10	All details from the employee table
3	6	DISTINCT department numbers: 10, 20, 30, 40, 50, 60
4	10	All department numbers (dept 20 appears three times)
5	7	Employee name and salary where salary is > 30000 rs
6	8	Employee name and hire date where city is not Delhi
7	4	Employee name and hire date where city is Delhi or Mumbai
8	5	Employee number and name where salary is between 30000 rs and 50000 rs
9	5	Employee number and name where salary is not between 30000 and 50000
10	8	All details where city is Delhi, Mumbai, Chennai, or Bangalore
11	4	All details where city is not Mumbai, Delhi, or Chennai
12	10	All details ordered by employee names descending
13	10	List of employees in ascending order
14	—	Table cse created (roll no, student name, city)
15	5	5 CSE students inserted
16	—	Table mechanical created with the same attributes
17	5	5 Mechanical students inserted (2 rows also in cse)
18	5	All details from the cse table
19	5	All details from the mechanical table
20	8	UNION — unique rows from either table (5 + 5 – 2)
21	10	UNION ALL — all rows including the 2 duplicates
22	2	INTERSECT — Amit Verma / Delhi and Sneha Reddy / Hyderabad

Question 3 (DISTINCT) collapses ten department values to six unique numbers; question 4 lists all ten. Questions 8 and 9 partition salaries with no overlap ($5 + 5 = 10$). Question 11 keeps Bangalore, Hyderabad, and Pune — cities that are not Mumbai, Delhi, or Chennai. Questions 20–22 show the set-operator identities on this seed: UNION has 8 rows, UNION ALL has 10, and INTERSECT has the 2 shared students.

9. Conclusion

A single SELECT can project columns, remove duplicates, filter with comparison and set predicates, and sort the result. DISTINCT answers “which unique values exist?” WHERE answers “which rows match this condition?” BETWEEN and IN express ranges and lists clearly. ORDER BY changes presentation only. UNION / UNION ALL / INTERSECT combine two compatible queries: unique rows from either side, every row including duplicates, or only rows common to both. Together these clauses are the core of everyday retrieval in SQL.

Ex – 5

Aggregates & Nested Queries

1. Aim

To create an employee table, insert 16 rows across four departments, and answer the lab questions with aggregate functions (SUM, AVG, MAX, MIN, COUNT), GROUP BY / HAVING, and nested queries (a scalar subquery, IN / NOT IN, EXISTS, and correlated subqueries, including the second-highest salary).

2. Theory

2.1 Aggregate functions

An aggregate function computes one value from many rows. COUNT(*) counts rows. SUM(salary) adds the salary column. AVG(salary) is the arithmetic mean. MAX and MIN return the largest and smallest values. Without GROUP BY, each aggregate is taken over the whole table and the query returns one row. SQLite prints this table's average as 56125.0 because AVG returns a real number even when the mean is a whole number of rupees.

2.2 GROUP BY

GROUP BY department makes one group per distinct department, then evaluates the aggregates inside each group. Every column in the SELECT list must be either grouped or aggregated. Question 7 therefore returns four rows (CSE, Civil, ECE, Mechanical), each with its own count, total, average, highest salary and lowest salary.

2.3 HAVING

WHERE filters rows before grouping. HAVING filters groups after the aggregates are computed, so it can test AVG(salary) or COUNT(*). WHERE AVG(salary) > 55000 is illegal. Question 8 keeps departments whose average is above 55000 (Civil at 45000.0 drops out). Question 9 keeps departments with more than three employees (Civil, with three, drops out).

2.4 Nested queries

A subquery is a SELECT inside another statement. A scalar subquery returns one value and can be compared with > or =. Question 10 compares each salary with the single overall average. An uncorrelated subquery does not refer to the outer query, so the engine can compute it once.

2.5 IN and NOT IN

IN (subquery) keeps an outer row when its value appears in the inner result. Question 13 keeps employees whose department is one of the departments that contain a Professor (CSE, Mechanical, ECE). NOT IN is the complement: question 14 keeps Civil, the only department with no Professor. If the subquery could return NULL, NOT IN would be unknown for every row and the result would be empty. Here department is NOT NULL, so that trap does not apply.

2.6 EXISTS and correlated subqueries

A correlated subquery mentions a column from the outer query, so it is re-evaluated for each outer row. EXISTS is true when the inner query returns at least one row; the selected columns do not matter, which is why question 15 selects the constant 1. It lists everyone who has a colleague in the same department on a higher salary — everyone except the highest-paid person in each department. Question 11 uses the same correlation to keep the row whose salary equals the department maximum. Question 16 compares each salary with the average of

that employee's own department, which is a different number from the overall average used in question 10.

2.7 Second-highest salary

Question 12 nests two scalar queries. The innermost `MAX(salary)` is the top salary (92000). The middle query takes the maximum salary that is strictly less than that value, which is the next distinct salary (90000). The outer query returns the employee who earns it. This pattern still means “second-highest distinct salary” if two people share the top salary. On this table Priya Nair is the only employee at 90000.

3. Schema

One table, `employee`, holds six attributes. `sr_no` is the employee number (primary key). `department` is stored as a name so `GROUP BY` results can be read without a second lookup. `salary` is a positive monthly amount in rupees.

Attribute	Type	Constraint	Role
sr_no	INTEGER	PRIMARY KEY	Employee number
employee_name	TEXT	NOT NULL	Employee name
job	TEXT	NOT NULL	Job title
department	TEXT	NOT NULL	Department name
salary	NUMERIC(10,2)	NOT NULL, CHECK > 0	Monthly salary (Rs)
city	TEXT	NOT NULL	Work city

4. Questions

Sixteen lab questions. Questions 1 and 2 build the table. Questions 3–6 are aggregates over every employee. Questions 7–9 group by department and, where asked, filter groups with `HAVING`. Questions 10–16 are nested queries: a comparison with the overall average, the highest salary in each department, the second-highest salary, `IN / NOT IN`, `EXISTS`, and a correlated department average.

1. Create a table `employee` with the following attributes:
 - sr_no
 - employee name
 - job
 - department
 - salary
 - city
2. Insert 16 rows in the above mentioned table.
Four departments: CSE (5), Mechanical (4), ECE (4) and Civil (3). Civil has no Professor, so the NOT IN question is non-empty. The highest salary (92000) and the second-highest salary (90000) are each earned by one employee.
3. Display the total number of employees from the employee table using `COUNT`.
4. Display the total salary of all employees from the employee table using `SUM`.
5. Display the average salary of all employees from the employee table using `AVG`.
6. Display the maximum and minimum salary from the employee table using `MAX` and `MIN`.
7. Display the number of employees, total salary, average salary, highest salary and lowest salary of each department using `COUNT`, `SUM`, `AVG`, `MAX`, `MIN` and `GROUP BY`.

8. Display the department and average salary of departments whose average salary is greater than 55000 using GROUP BY and HAVING.
9. Display the department and the number of employees for departments that have more than 3 employees using GROUP BY and HAVING.
10. Display the employee name, department and salary of employees who earn more than the average salary, using a nested query.
11. Display the employee with the highest salary in each department using a subquery.
12. Display the employee name, department and salary of the employee who earns the second-highest salary, using a nested query.
13. Display employees who work in a department that has a Professor, using IN and a subquery.
14. Display employees who do not work in a department that has a Professor, using NOT IN and a subquery.
department is NOT NULL, so the subquery cannot return NULL. A NULL in a NOT IN list would make the predicate unknown for every row.
15. Display employees for whom there exists a higher-paid colleague in the same department, using EXISTS.
16. Display employees who earn more than the average salary of their own department, using a correlated subquery.
The comparison is strict. Rahul Das earns exactly the Civil average (45000), so this question does not list him.

5. Procedure

Drop employee if it already exists so the script can be re-run. Create the table, insert the 16 rows, then run each query in order. Grouped and nested results are ordered so the same script prints the same rows on every run. The SQL for each question:

Q. No.	SQL solution
1	<pre>CREATE TABLE employee (sr_no INTEGER PRIMARY KEY, employee_name TEXT NOT NULL, job TEXT NOT NULL, department TEXT NOT NULL, salary NUMERIC(10, 2) NOT NULL CHECK (salary > 0), city TEXT NOT NULL);</pre>
2	<pre>INSERT INTO employee (sr_no, employee_name, job, department, salary, city) VALUES (1, 'Rajesh Kumar', 'Professor', 'CSE', 92000.00, 'Delhi'), (2, 'Ananya Sharma', 'Associate Professor', 'CSE', 78000.00, 'Bangalore'), (3, 'Fatima Khan', 'Assistant Professor', 'CSE', 56000.00, 'Bangalore'), (4, 'Karan Iyer', 'Lab Instructor', 'CSE', 42000.00, 'Mumbai'), (5, 'Dev Patel', 'Teaching Assistant', 'CSE', 28000.00, 'Ahmedabad'), (6, 'Vikram Singh', 'Professor', 'Mechanical', 88000.00, 'Hyderabad'), (7, 'Rohan Mehta', 'Associate Professor', 'Mechanical', 64000.00, 'Mumbai'), (8, 'Arjun Patel', 'Assistant Professor', 'Mechanical', 48000.00, 'Pune'), (9, 'Mohit Jain', 'Lab Instructor', 'Mechanical', 32000.00, 'Jaipur'), (10, 'Priya Nair', 'Professor', 'ECE', 90000.00, 'Chennai'), (11, 'Neha Gupta', 'Associate Professor', 'ECE', 58000.00, 'Delhi'), (12, 'Meera Joshi', 'Assistant Professor', 'ECE', 51000.00, 'Chennai'), (13, 'Kavya Menon', 'Lab Instructor', 'ECE', 36000.00, 'Kochi'), (14, 'Sneha Reddy', 'Associate Professor', 'Civil', 60000.00, 'Hyderabad'), (15, 'Rahul Das', 'Assistant Professor', 'Civil', 45000.00, 'Kolkata'), (16, 'Tushar Rao', 'Lab Instructor', 'Civil', 30000.00, 'Nagpur');</pre>

Q. No.	SQL solution
3	SELECT COUNT(*) AS employee_count FROM employee;
4	SELECT SUM(salary) AS total_salary FROM employee;
5	SELECT AVG(salary) AS average_salary FROM employee;
6	SELECT MAX(salary) AS maximum_salary, MIN(salary) AS minimum_salary FROM employee;
7	SELECT department, COUNT(*) AS employee_count, SUM(salary) AS total_salary, AVG(salary) AS average_salary, MAX(salary) AS highest_salary, MIN(salary) AS lowest_salary FROM employee GROUP BY department ORDER BY department;
8	SELECT department, AVG(salary) AS average_salary FROM employee GROUP BY department HAVING AVG(salary) > 55000 ORDER BY department;
9	SELECT department, COUNT(*) AS employee_count FROM employee GROUP BY department HAVING COUNT(*) > 3 ORDER BY department;
10	SELECT employee_name, department, salary FROM employee WHERE salary > (SELECT AVG(salary) FROM employee) ORDER BY salary DESC;
11	SELECT e.employee_name, e.department, e.salary FROM employee AS e WHERE e.salary = (SELECT MAX(c.salary) FROM employee AS c WHERE c.department = e.department) ORDER BY e.department;
12	SELECT employee_name, department, salary AS second_highest_salary FROM employee WHERE salary = (SELECT MAX(salary) FROM employee WHERE salary < (SELECT MAX(salary) FROM employee));
13	SELECT employee_name, job, department, salary FROM employee WHERE department IN (SELECT department FROM employee WHERE job = 'Professor') ORDER BY department, employee_name;
14	SELECT employee_name, job, department, salary FROM employee WHERE department NOT IN (SELECT department FROM employee WHERE job = 'Professor') ORDER BY department, employee_name;

Q. No.	SQL solution
15	<pre> SELECT e.employee_name, e.department, e.salary FROM employee AS e WHERE EXISTS (SELECT 1 FROM employee AS c WHERE c.department = e.department AND c.salary > e.salary) ORDER BY e.department, e.salary DESC; </pre>
16	<pre> SELECT e.employee_name, e.department, e.salary FROM employee AS e WHERE e.salary > (SELECT AVG(d.salary) FROM employee AS d WHERE d.department = e.department) ORDER BY e.department, e.salary DESC; </pre>

6. Source Code

SQLite script used in the lab compiler (30-09-2026/employee.sql). For the class SQL Server, run employee.sqlserver.sql.

```

-- =====
-- DBMS Lab · 30-09-2026
-- Employee table: CREATE, seed 16 rows, then aggregate functions
-- and nested queries.
-- SUM, AVG, MAX, MIN, COUNT, GROUP BY, HAVING
-- Subqueries: scalar, IN, NOT IN, EXISTS, correlated
-- SQLite (sql.js compiler / DB Browser / sqlite3)
-- For the class SQL Server, run employee.sqlserver.sql instead.
--
-- Lab questions
-- 1. Create a table employee with the following attributes:
--   sr no, employee name, job, department, salary, city
-- 2. Insert 16 rows in the above mentioned table
-- 3. Display the total number of employees using COUNT
-- 4. Display the total salary of all employees using SUM
-- 5. Display the average salary of all employees using AVG
-- 6. Display the maximum and minimum salary using MAX and MIN
-- 7. For each department, display COUNT, SUM, AVG, MAX and MIN
--   using GROUP BY
-- 8. Display departments whose average salary is greater than
--   55000 using GROUP BY and HAVING
-- 9. Display departments that have more than 3 employees
--   using GROUP BY and HAVING
-- 10. Display employees who earn more than the average salary
--   (nested query)
-- 11. Display the employee with the highest salary in each
--   department using a subquery
-- 12. Display the employee who earns the second-highest salary
--   using a nested query
-- 13. Display employees who work in a department that has a
--   Professor, using IN and a subquery
-- 14. Display employees who do not work in a department that
--   has a Professor, using NOT IN and a subquery
-- 15. Display employees for whom there EXISTS a higher-paid
--   colleague in the same department (correlated)
-- 16. Display employees who earn more than the average salary
--   of their own department (correlated subquery)
-- =====

DROP TABLE IF EXISTS employee;

-- -----
-- 1. Create a table employee with the following attributes:
--   sr no, employee name, job, department, salary, city
-- -----
CREATE TABLE employee (
  sr_no          INTEGER PRIMARY KEY,
  employee_name  TEXT NOT NULL,
  job            TEXT NOT NULL,
  department     TEXT NOT NULL,
  salary        NUMERIC(10, 2) NOT NULL CHECK (salary > 0),
  city          TEXT NOT NULL
);

-- -----
-- 2. Insert 16 rows in the above mentioned table
--   · four departments: CSE (5), Mechanical (4), ECE (4), Civil (3)
--   · Civil has no Professor, so NOT IN is non-empty
--   · salaries vary so SUM, AVG, MAX, MIN and HAVING differ
--   · the overall maximum (92000) and second maximum (90000)

```

```
-- are each earned by exactly one employee
-----
INSERT INTO employee
(sr_no, employee_name, job, department, salary, city)
VALUES
(1, 'Rajesh Kumar', 'Professor', 'CSE', 92000.00, 'Delhi'),
(2, 'Ananya Sharma', 'Associate Professor', 'CSE', 78000.00, 'Bangalore'),
(3, 'Fatima Khan', 'Assistant Professor', 'CSE', 56000.00, 'Bangalore'),
(4, 'Karan Iyer', 'Lab Instructor', 'CSE', 42000.00, 'Mumbai'),
(5, 'Dev Patel', 'Teaching Assistant', 'CSE', 28000.00, 'Ahmedabad'),
(6, 'Vikram Singh', 'Professor', 'Mechanical', 88000.00, 'Hyderabad'),
(7, 'Rohan Mehta', 'Associate Professor', 'Mechanical', 64000.00, 'Mumbai'),
(8, 'Arjun Patel', 'Assistant Professor', 'Mechanical', 48000.00, 'Pune'),
(9, 'Mohit Jain', 'Lab Instructor', 'Mechanical', 32000.00, 'Jaipur'),
(10, 'Priya Nair', 'Professor', 'ECE', 90000.00, 'Chennai'),
(11, 'Neha Gupta', 'Associate Professor', 'ECE', 58000.00, 'Delhi'),
(12, 'Meera Joshi', 'Assistant Professor', 'ECE', 51000.00, 'Chennai'),
(13, 'Kavya Menon', 'Lab Instructor', 'ECE', 36000.00, 'Kochi'),
(14, 'Sneha Reddy', 'Associate Professor', 'Civil', 60000.00, 'Hyderabad'),
(15, 'Rahul Das', 'Assistant Professor', 'Civil', 45000.00, 'Kolkata'),
(16, 'Tushar Rao', 'Lab Instructor', 'Civil', 30000.00, 'Nagpur');
```

```
-- 3. Display the total number of employees from the employee
-- table using COUNT
-----
```

```
-- Total number of employees
SELECT COUNT(*) AS employee_count
FROM employee;
```

```
-- 4. Display the total salary of all employees from the
-- employee table using SUM
-----
```

```
-- Total salary of all employees
SELECT SUM(salary) AS total_salary
FROM employee;
```

```
-- 5. Display the average salary of all employees from the
-- employee table using AVG
-----
```

```
-- Average salary of all employees
SELECT AVG(salary) AS average_salary
FROM employee;
```

```
-- 6. Display the maximum and minimum salary from the employee
-- table using MAX and MIN
-----
```

```
-- Maximum and minimum salary
SELECT MAX(salary) AS maximum_salary,
       MIN(salary) AS minimum_salary
FROM employee;
```

```
-- 7. Display the number of employees, total salary, average
-- salary, highest salary and lowest salary of each
-- department using COUNT, SUM, AVG, MAX, MIN and GROUP BY
-----
```

```
-- Aggregates for each department
SELECT department,
       COUNT(*) AS employee_count,
       SUM(salary) AS total_salary,
       AVG(salary) AS average_salary,
       MAX(salary) AS highest_salary,
       MIN(salary) AS lowest_salary
FROM employee
GROUP BY department
ORDER BY department;
```

```
-- 8. Display the department and average salary of departments
-- whose average salary is greater than 55000 using
-- GROUP BY and HAVING
-----
```

```
-- Departments with average salary above 55000
SELECT department,
       AVG(salary) AS average_salary
FROM employee
GROUP BY department
HAVING AVG(salary) > 55000
ORDER BY department;
```

```
-- 9. Display the department and the number of employees for
-- departments that have more than 3 employees using
-- GROUP BY and HAVING
-----
```

```
-- Departments with more than 3 employees
SELECT department,
       COUNT(*) AS employee_count
```

```

FROM employee
GROUP BY department
HAVING COUNT(*) > 3
ORDER BY department;

-- -----
-- 10. Display the employee name, department and salary of
--     employees who earn more than the average salary,
--     using a nested query
-- -----
-- Employees earning more than the average salary
SELECT employee_name, department, salary
FROM employee
WHERE salary > (SELECT AVG(salary) FROM employee)
ORDER BY salary DESC;

-- -----
-- 11. Display the employee with the highest salary in each
--     department using a subquery
-- -----
-- Highest salary in each department
SELECT e.employee_name, e.department, e.salary
FROM employee AS e
WHERE e.salary = (
    SELECT MAX(c.salary)
    FROM employee AS c
    WHERE c.department = e.department
)
ORDER BY e.department;

-- -----
-- 12. Display the employee name, department and salary of the
--     employee who earns the second-highest salary, using a
--     nested query
-- -----
-- Second-highest salary
SELECT employee_name, department, salary AS second_highest_salary
FROM employee
WHERE salary = (
    SELECT MAX(salary)
    FROM employee
    WHERE salary < (SELECT MAX(salary) FROM employee)
);

-- -----
-- 13. Display employees who work in a department that has a
--     Professor, using IN and a subquery
-- -----
-- Employees in a department that has a Professor
SELECT employee_name, job, department, salary
FROM employee
WHERE department IN (
    SELECT department
    FROM employee
    WHERE job = 'Professor'
)
ORDER BY department, employee_name;

-- -----
-- 14. Display employees who do not work in a department that
--     has a Professor, using NOT IN and a subquery
-- -----
-- Employees not in a department that has a Professor
SELECT employee_name, job, department, salary
FROM employee
WHERE department NOT IN (
    SELECT department
    FROM employee
    WHERE job = 'Professor'
)
ORDER BY department, employee_name;

-- -----
-- 15. Display employees for whom there EXISTS a higher-paid
--     colleague in the same department (correlated subquery)
-- -----
-- Employees with a higher-paid colleague
SELECT e.employee_name, e.department, e.salary
FROM employee AS e
WHERE EXISTS (
    SELECT 1
    FROM employee AS c
    WHERE c.department = e.department
    AND c.salary > e.salary
)
ORDER BY e.department, e.salary DESC;

-- -----
-- 16. Display employees who earn more than the average salary
--     of their own department (correlated subquery)
-- -----
-- Employees above their department average

```

```

SELECT e.employee_name, e.department, e.salary
FROM employee AS e
WHERE e.salary > (
    SELECT AVG(d.salary)
    FROM employee AS d
    WHERE d.department = e.department
)
ORDER BY e.department, e.salary DESC;

```

7. Output

The script was executed in SQLite in memory. CREATE prints a status line and INSERT prints the row count. Each SELECT prints the result set that SQLite returned. Whole-number salaries print as integers; AVG prints a real (56125.0, 59200.0, 58750.0, and so on).

```

SQL> DROP TABLE IF EXISTS employee;
-- OK

SQL> CREATE TABLE employee (
    sr_no          INTEGER PRIMARY KEY,
    employee_name  TEXT NOT NULL,
    job            TEXT NOT NULL,
    department     TEXT NOT NULL,
    salary         NUMERIC(10, 2) NOT NULL CHECK (salary > 0),
    city          TEXT NOT NULL
);
-- OK

SQL> INSERT INTO employee
(sr_no, employee_name, job, department, salary, city)
VALUES
(1, 'Rajesh Kumar', 'Professor', 'CSE', 92000.00, 'Delhi'),
(2, 'Ananya Sharma', 'Associate Professor', 'CSE', 78000.00, 'Bangalore'),
(3, 'Fatima Khan', 'Assistant Professor', 'CSE', 56000.00, 'Bangalore'),
(4, 'Karan Iyer', 'Lab Instructor', 'CSE', 42000.00, 'Mumbai'),
(5, 'Dev Patel', 'Teaching Assistant', 'CSE', 28000.00, 'Ahmedabad'),
(6, 'Vikram Singh', 'Professor', 'Mechanical', 88000.00, 'Hyderabad'),
(7, 'Rohan Mehta', 'Associate Professor', 'Mechanical', 64000.00, 'Mumbai'),
(8, 'Arjun Patel', 'Assistant Professor', 'Mechanical', 48000.00, 'Pune'),
(9, 'Mohit Jain', 'Lab Instructor', 'Mechanical', 32000.00, 'Jaipur'),
(10, 'Priya Nair', 'Professor', 'ECE', 90000.00, 'Chennai'),
(11, 'Neha Gupta', 'Associate Professor', 'ECE', 58000.00, 'Delhi'),
(12, 'Meera Joshi', 'Assistant Professor', 'ECE', 51000.00, 'Chennai'),
(13, 'Kavya Menon', 'Lab Instructor', 'ECE', 36000.00, 'Kochi'),
(14, 'Sneha Reddy', 'Associate Professor', 'Civil', 60000.00, 'Hyderabad'),
(15, 'Rahul Das', 'Assistant Professor', 'Civil', 45000.00, 'Kolkata'),
(16, 'Tushar Rao', 'Lab Instructor', 'Civil', 30000.00, 'Nagpur');
-- 16 row(s) inserted

SQL> SELECT COUNT(*) AS employee_count
FROM employee;
employee_count
-----
16

SQL> SELECT SUM(salary) AS total_salary
FROM employee;
total_salary
-----
898000

SQL> SELECT AVG(salary) AS average_salary
FROM employee;
average_salary
-----
56125.0

SQL> SELECT MAX(salary) AS maximum_salary,
MIN(salary) AS minimum_salary
FROM employee;
maximum_salary  minimum_salary
-----
92000           28000

SQL> SELECT department,
COUNT(*) AS employee_count,
SUM(salary) AS total_salary,
AVG(salary) AS average_salary,
MAX(salary) AS highest_salary,
MIN(salary) AS lowest_salary
FROM employee
GROUP BY department
ORDER BY department;
department  employee_count  total_salary  average_salary  highest_salary  lowest_salary
-----
CSE         5              296000       59200.0         92000          28000
Civil       3              135000       45000.0         60000          30000
ECE         4              235000       58750.0         90000          36000
Mechanical  4              232000       58000.0         88000          32000

SQL> SELECT department,
AVG(salary) AS average_salary
FROM employee
GROUP BY department
HAVING AVG(salary) > 55000
ORDER BY department;
department  average_salary
-----

```

```
CSE      59200.0
ECE      58750.0
Mechanical 58000.0
```

```
SQL> SELECT department,
COUNT(*) AS employee_count
FROM employee
GROUP BY department
HAVING COUNT(*) > 3
ORDER BY department;
department employee_count
-----
CSE      5
ECE      4
Mechanical 4
```

```
SQL> SELECT employee_name, department, salary
FROM employee
WHERE salary > (SELECT AVG(salary) FROM employee)
ORDER BY salary DESC;
employee_name department salary
-----
Rajesh Kumar CSE      92000
Priya Nair ECE      90000
Vikram Singh Mechanical 88000
Ananya Sharma CSE      78000
Rohan Mehta Mechanical 64000
Sneha Reddy Civil     60000
Neha Gupta ECE      58000
```

```
SQL> SELECT e.employee_name, e.department, e.salary
FROM employee AS e
WHERE e.salary = (
SELECT MAX(c.salary)
FROM employee AS c
WHERE c.department = e.department
)
ORDER BY e.department;
employee_name department salary
-----
Rajesh Kumar CSE      92000
Sneha Reddy Civil     60000
Priya Nair ECE      90000
Vikram Singh Mechanical 88000
```

```
SQL> SELECT employee_name, department, salary AS second_highest_salary
FROM employee
WHERE salary = (
SELECT MAX(salary)
FROM employee
WHERE salary < (SELECT MAX(salary) FROM employee)
);
employee_name department second_highest_salary
-----
Priya Nair ECE      90000
```

```
SQL> SELECT employee_name, job, department, salary
FROM employee
WHERE department IN (
SELECT department
FROM employee
WHERE job = 'Professor'
)
ORDER BY department, employee_name;
employee_name job department salary
-----
Ananya Sharma Associate Professor CSE      78000
Dev Patel Teaching Assistant CSE      28000
Fatima Khan Assistant Professor CSE      56000
Karan Iyer Lab Instructor CSE      42000
Rajesh Kumar Professor CSE      92000
Kavya Menon Lab Instructor ECE      36000
Meera Joshi Assistant Professor ECE      51000
Neha Gupta Associate Professor ECE      58000
Priya Nair Professor ECE      90000
Arjun Patel Assistant Professor Mechanical 48000
Mohit Jain Lab Instructor Mechanical 32000
Rohan Mehta Associate Professor Mechanical 64000
Vikram Singh Professor Mechanical 88000
```

```
SQL> SELECT employee_name, job, department, salary
FROM employee
WHERE department NOT IN (
SELECT department
FROM employee
WHERE job = 'Professor'
)
ORDER BY department, employee_name;
employee_name job department salary
-----
Rahul Das Assistant Professor Civil     45000
Sneha Reddy Associate Professor Civil     60000
Tushar Rao Lab Instructor Civil     30000
```

```
SQL> SELECT e.employee_name, e.department, e.salary
FROM employee AS e
WHERE EXISTS (
SELECT 1
FROM employee AS c
WHERE c.department = e.department
AND c.salary > e.salary
)
ORDER BY e.department, e.salary DESC;
employee_name department salary
-----
```

Ananya Sharma	CSE	78000
Fatima Khan	CSE	56000
Karan Iyer	CSE	42000
Dev Patel	CSE	28000
Rahul Das	Civil	45000
Tushar Rao	Civil	30000
Neha Gupta	ECE	58000
Meera Joshi	ECE	51000
Kavya Menon	ECE	36000
Rohan Mehta	Mechanical	64000
Arjun Patel	Mechanical	48000
Mohit Jain	Mechanical	32000

```
SQL> SELECT e.employee_name, e.department, e.salary
FROM employee AS e
WHERE e.salary > (
    SELECT AVG(d.salary)
    FROM employee AS d
    WHERE d.department = e.department
)
ORDER BY e.department, e.salary DESC;
employee_name department salary
-----
Rajesh Kumar    CSE          92000
Ananya Sharma   CSE          78000
Sneha Reddy     Civil         60000
Priya Nair      ECE          90000
Vikram Singh    Mechanical    88000
Rohan Mehta     Mechanical    64000
```

8. Results

Sixteen employees load successfully. Every question returns a non-empty result on this seed. The figures below are the row counts and values from the SQLite run in section 7.

Q. No.	Rows	What the result shows
1	—	Table employee created with 6 attributes
2	16	16 rows inserted across CSE, Mechanical, ECE and Civil
3	1	COUNT(*) = 16 employees
4	1	SUM(salary) = 898000
5	1	AVG(salary) = 56125.0
6	1	MAX = 92000 and MIN = 28000
7	4	COUNT, SUM, AVG, MAX and MIN for each of the 4 departments
8	3	CSE (59200.0), ECE (58750.0), Mechanical (58000.0); Civil excluded
9	3	CSE (5), ECE (4), Mechanical (4); Civil has only 3
10	7	Names earning more than the overall average 56125.0
11	4	Rajesh Kumar, Sneha Reddy, Priya Nair, Vikram Singh
12	1	Second-highest salary: Priya Nair, ECE, 90000
13	13	IN: employees in CSE, Mechanical or ECE (a Professor exists)
14	3	NOT IN: Rahul Das, Sneha Reddy and Tushar Rao (Civil)
15	12	EXISTS: everyone except the 4 department-wise top salaries
16	6	Correlated: salary greater than that department's own average

Department totals from question 7 add up to the overall total: 296000 + 135000 + 235000 + 232000 = 898000, and 5 + 3 + 4 + 4 = 16. Question 10 returns seven employees above the company average of 56125.0. Question 16 returns six employees above their own department average. Neha Gupta (ECE, 58000) is in the first list only: 58000 is above 56125.0, but the ECE average is 58750.0. Fatima Khan (56000) is below both the company average and the CSE average (59200.0). Rahul Das earns exactly the Civil average (45000), so question 16 excludes him, and 45000 is also below the company average.

Question 12 returns one row because 90000 occurs once. Question 15 returns 12 rows: 16 employees minus the four people who have no higher-paid colleague in their department, the same four names question 11 returns.

9. Conclusion

Aggregate functions collapse many rows into one value, and GROUP BY repeats that collapse once per group. HAVING then keeps or drops whole groups. A nested query uses one result as the input of another: a scalar subquery for a comparison, IN / NOT IN for membership, and EXISTS for a yes-or-no test. When the inner query refers to the outer row it is correlated, so each employee can be compared with their own department rather than with the whole company. Together these are the usual tools for summary questions and for questions of the form “more than the average” or “the highest in each group”.